The Voice of Monetary Policy

Yuriy Gorodnichenko, Tho Pham, and Oleksandr Talavera

American Economic Review, 2023, 113(2): 548-584

Prepared for ISEG Summer School 2026, by Benjamin W. Arold

Motivation & Research Question

- Central banks increasingly steer expectations through communication (e.g., forward guidance) -- but are they using it to full potential?
- Mehrabian's 7-38-55 rule: words carry ~7% of a message, tone of voice ~38%, body language ~55%. Yet research studies only the TEXT of statements and minutes.
- Research question: after controlling for the Fed's ACTIONS and the TEXT SENTIMENT of its messages, does the TONE OF VOICE of the Fed chair move financial markets?
  - Focus on the Q&A of post-FOMC press conferences -- the only unscripted part, where soft information can leak through vocal cues.
- First study to measure central-bank communication through the vocal (nonverbal) dimension.

Data & Setting

- Sample: April 2011 (first FOMC press conference) to June 2019 -- 68 FOMC meetings, 36 press conferences, 3 chairs (Bernanke, Yellen, Powell).
- Press-conference videos from the Fed's official YouTube channel; only the AUDIO track is used (consistent across speakers and camera angles).
- Each conference split manually into audio segments = one chair answer per question: 692 answers, avg 19 per conference, mostly 1-3 minutes.
- Fed ACTIONS controlled for with Swanson (2021) policy shocks (FFR, forward guidance, asset purchase) and the Wu-Xia (2016) shadow rate.
- Daily financial outcomes from ETFs: SPY (S&P 500), VIX/VIXY, GOVT, LQD, GLD, and dollar/euro and dollar/yen exchange rates.

Method (1): Detecting Emotions from the Chair's Voice

- Goal: build a deep-learning classifier that maps sound waves into emotions.
- Vocal features extracted with Librosa: 128 mel-spectrogram frequencies + 40 MFCCs + 12 chroma coefficients = 180 inputs per audio clip.
- Training data: RAVDESS and TESS acted-emotion speech databases; keep 5 emotions (happy, pleasantly surprised, neutral, sad, angry); 80/20 train/test.
- Model: fully-connected neural net in Keras/TensorFlow -- 180 -> 200 -> 200 -> 200 nodes, softmax 5-way output, dropout 0.3 to limit overfitting.
- Overall test accuracy 84% (74-87% per emotion). Customized model (vs commercial software) allows tuning of emotions and parameters -- objective and reproducible.

Method (2): VoiceTone and the Text-Sentiment Control

- Each answer is labeled positive (happy/surprised), negative (sad/angry), or neutral.
- VoiceTone = (Positive - Negative) / (Positive + Negative), aggregated per conference; ranges from -1 to +1.
- Text sentiment must be netted out: BERT contextual, bidirectional word embeddings feed a bidirectional-LSTM net, fine-tuned on 1997-2010 FOMC statements hand-scored from -10 (hawkish) to +10 (dovish).
- TextSentiment = (Dovish - Hawkish) / (Dovish + Hawkish); 81% accuracy; robust to RoBERTa, FinBERT, dictionary, and human coding.
- Face validity: Bernanke's defensive Sept-2013 'taper-tantrum' conference scores near -1; Powell's dovish surprise (Mar-2019) scores +0.92.

Descriptives: Voice Tone and Text Sentiment by Chair

TABLE 1—FOMC MEETING STATISTICS

	All (1)	Bernanke (2)	Yellen (3)	Powell (4)
Meetings	68	25	32	11
Press conferences	36	12	16	8
Panel A. Voice analysis of responses in Q&A during press conferences				
Answers (count)				
Positive	377	200	109	68
Negative	285	43	131	111
Neutral	30	0	28	2
Voice tone				
Mean	0.09	0.64	−0.13	−0.30
Standard deviation	0.75	0.58	0.61	0.82
Panel B. Textual analysis				
Statement				
Hawkish paragraphs	37	8	20	9
Dovish paragraphs	223	105	108	10
Neutral paragraphs	64	6	37	21
Text sentiment				
Mean	0.66	0.86	0.71	0.09
Standard deviation	0.53	0.27	0.34	0.94
Remarks				
Hawkish paragraphs	83	15	47	21
Dovish paragraphs	244	97	106	41
Neutral paragraphs	119	41	50	28
Text sentiment				
Mean	0.50	0.75	0.38	0.35
Standard deviation	0.37	0.24	0.33	0.47
Q&A				
Hawkish answers	233	77	95	61
Dovish answers	339	137	120	82
Neutral answers	120	29	53	38
Text sentiment				
Mean	0.18	0.29	0.11	0.14
Standard deviation	0.30	0.30	0.26	0.35
Statement, Remarks, Q&A				
Text sentiment				
Mean	0.52	0.71	0.50	0.15
Standard deviation	0.40	0.30	0.35	0.52

Note: FOMC meeting statistics, 2011-2019. Panel A: counts of positive/negative/neutral answers and the resulting VoiceTone (eq. 2). Bernanke sounds most positive (mean 0.64), Powell most negative (-0.30), with large within-speaker variation. Panel B: text sentiment (eq. 3). (Table 1)

Voice Tone Carries Independent Information

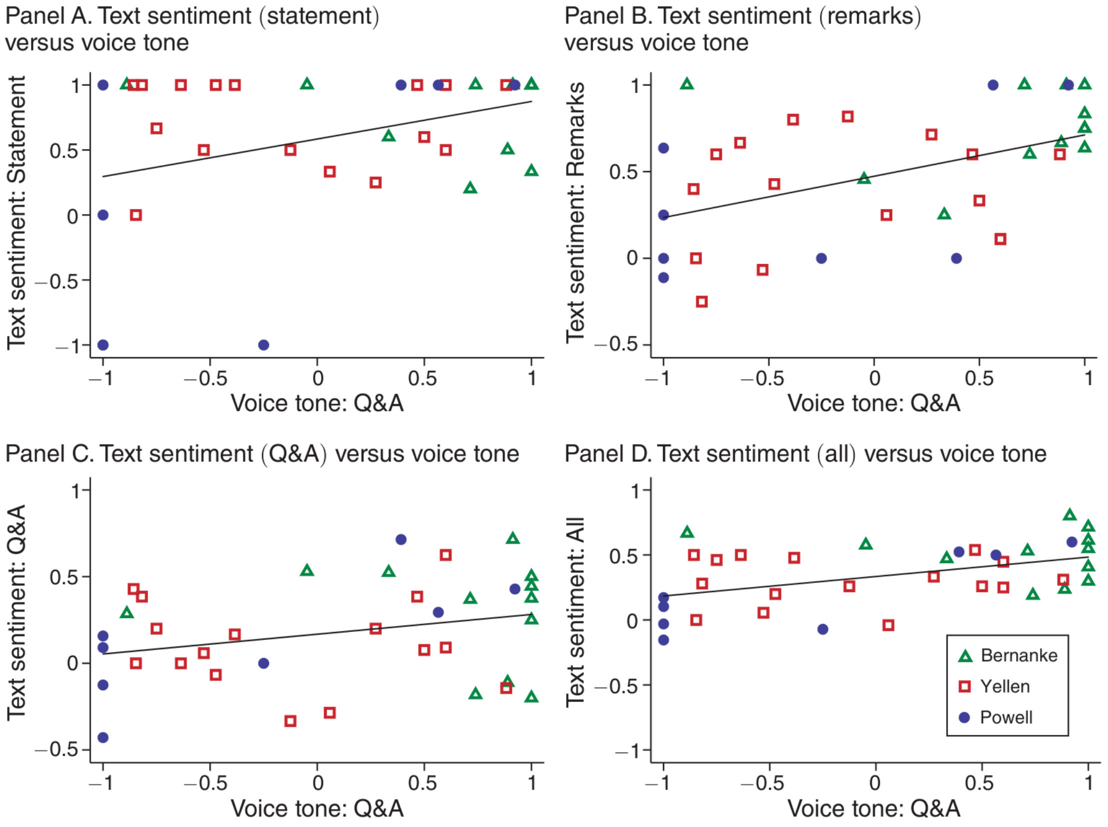

FIGURE 1. VOICE TONE VERSUS TEXT SENTIMENT

Note: This figure shows the joint distribution of voice tone and text sentiment across FOMC meetings.

Note: Joint distribution of Q&A voice tone and text sentiment across FOMC meetings, by chair. (Figure 1)

- Voice tone and text sentiment are congruent but far from identical (corr = 0.37 with statement sentiment).
- Voice tone is only weakly correlated with policy shocks (|corr| <= 0.23) -- so it provides independent variation.
- Identification: daily local projections (Jorda 2005) regress each outcome on VoiceTone, controlling for TextSentiment, FFR/FG/AP shocks, and the shadow rate.
- Coefficients plotted over horizons h = 0..15 days; 90% bias-corrected bootstrap CIs (small sample, 68 meetings).

Headline Result: A Positive Voice Tone Lifts Share Prices

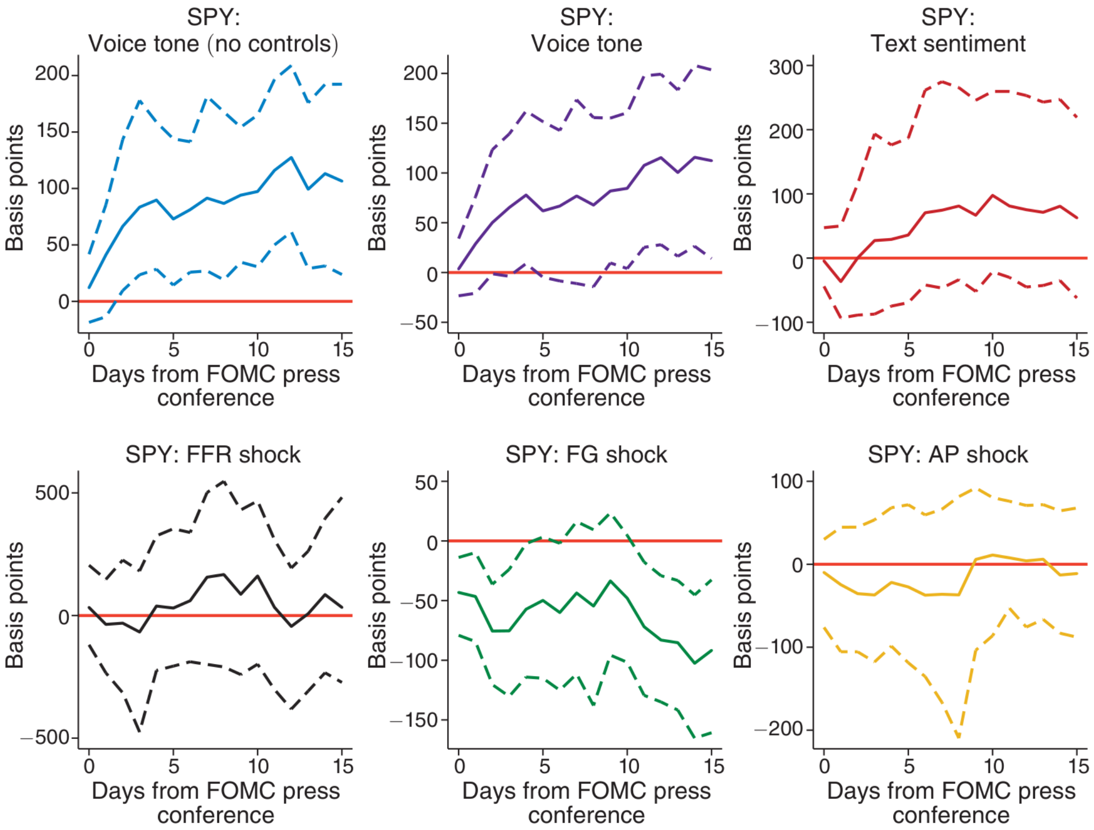

FIGURE 3. RESPONSE OF SPY ETF (S&P 500) TO POLICY ACTIONS AND MESSAGES

Note: Response of SPY (S&P 500 ETF) to a unit increase in each factor. Top-center: with full controls, a more positive voice tone raises returns ~100 bp after five days (a +1 SD tone ~ +75 bp), significant at 10%. The effect rivals a 1 SD forward-guidance shock and contributes ~20% of the R-squared. (Figure 3)

A Positive Tone Also Lowers Expected Volatility

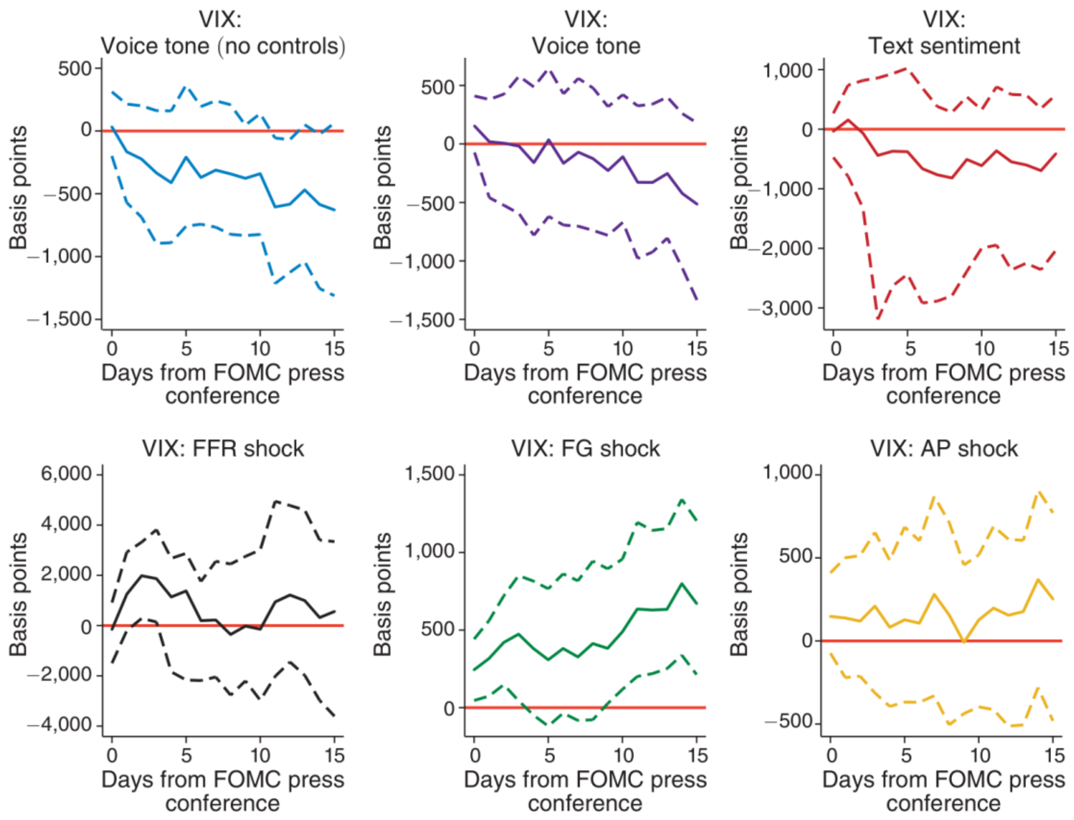

FIGURE 4. RESPONSE OF VIX (CBOE VOLATILITY INDEX) TO POLICY ACTIONS AND MESSAGES

Notes: This figure reports the estimated slope coefficients b (specification (4)) for policy communication/actions.

Note: Response of the VIX to policy actions and messages. A more positive voice tone (and more dovish text) reduces current and anticipated volatility, while FFR/FG/AP action shocks raise it. Voice tone similarly lowers interest-rate risk and expected inflation, and appreciates the dollar vs the euro. (Figure 4)

Intraday Evidence: Markets React Within Minutes

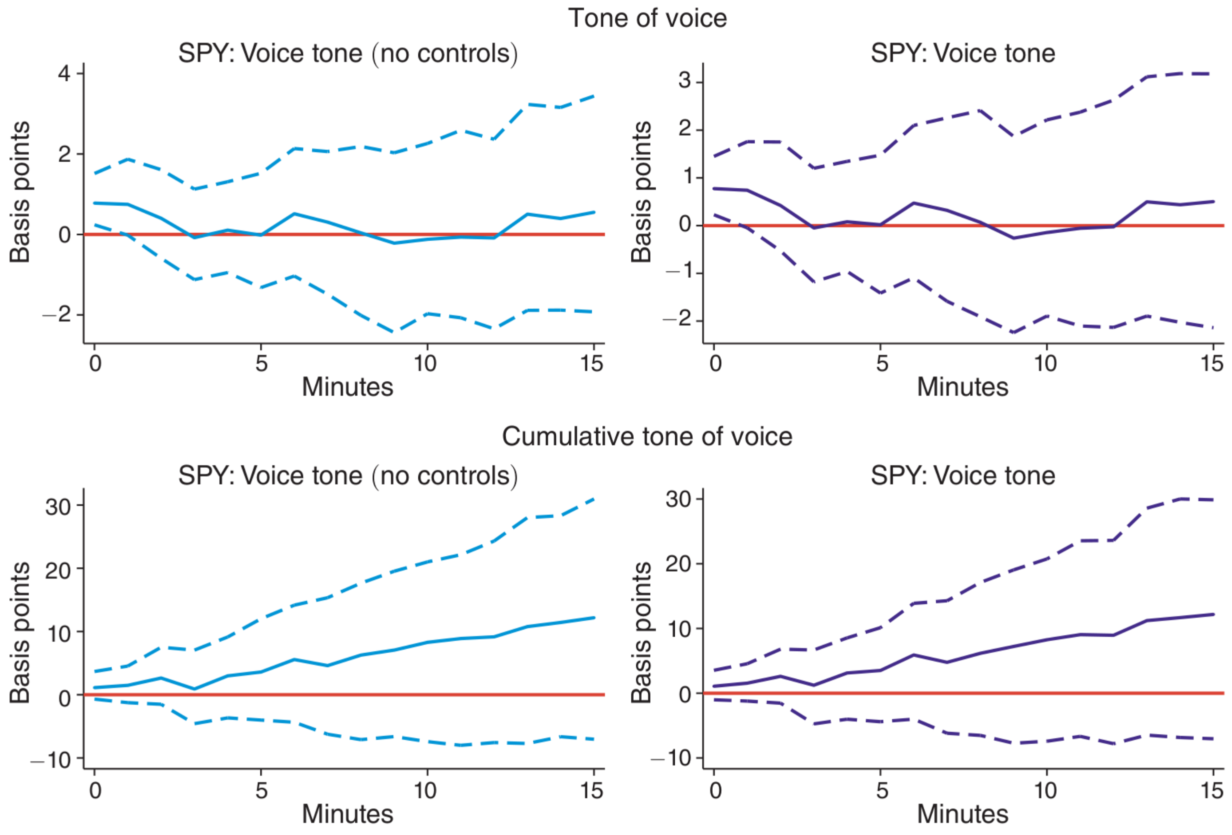

FIGURE 15. INTRADAY RESPONSES OF STOCK PRICES (SPY) TO VOICE TONE

Note: Intraday SPY response to voice tone, answer by answer (top) and cumulatively over the press conference (bottom), with meeting and question-order fixed effects. Impact effect is small (~1 bp) but positive; the cumulative response grows as the tone becomes clearer -- markets need time to process nonverbal cues. (Figure 15)

Takeaways & Implications

- After netting out Fed actions and text sentiment, a more positive vocal tone significantly RAISES stock prices: a +1 SD tone adds ~75 bp to the S&P 500 over ~5 days -- comparable to a 1 SD forward-guidance shock.
- Voice tone also lowers expected volatility, reduces interest-rate risk and expected inflation, and appreciates the dollar vs the euro; the bond-market response is weak and mixed.
- Voice tone accounts for ~20% of the explained variation -- economically as important as forward-guidance shocks.
- Robust to alternative text measures (RoBERTa, FinBERT, dictionary, human coding), macro-surprise and media controls, and chair fixed effects.
- Bottom line: HOW policy is communicated -- not only WHAT is said -- moves markets. Nonverbal delivery is a genuine, under-used central-bank communication tool.